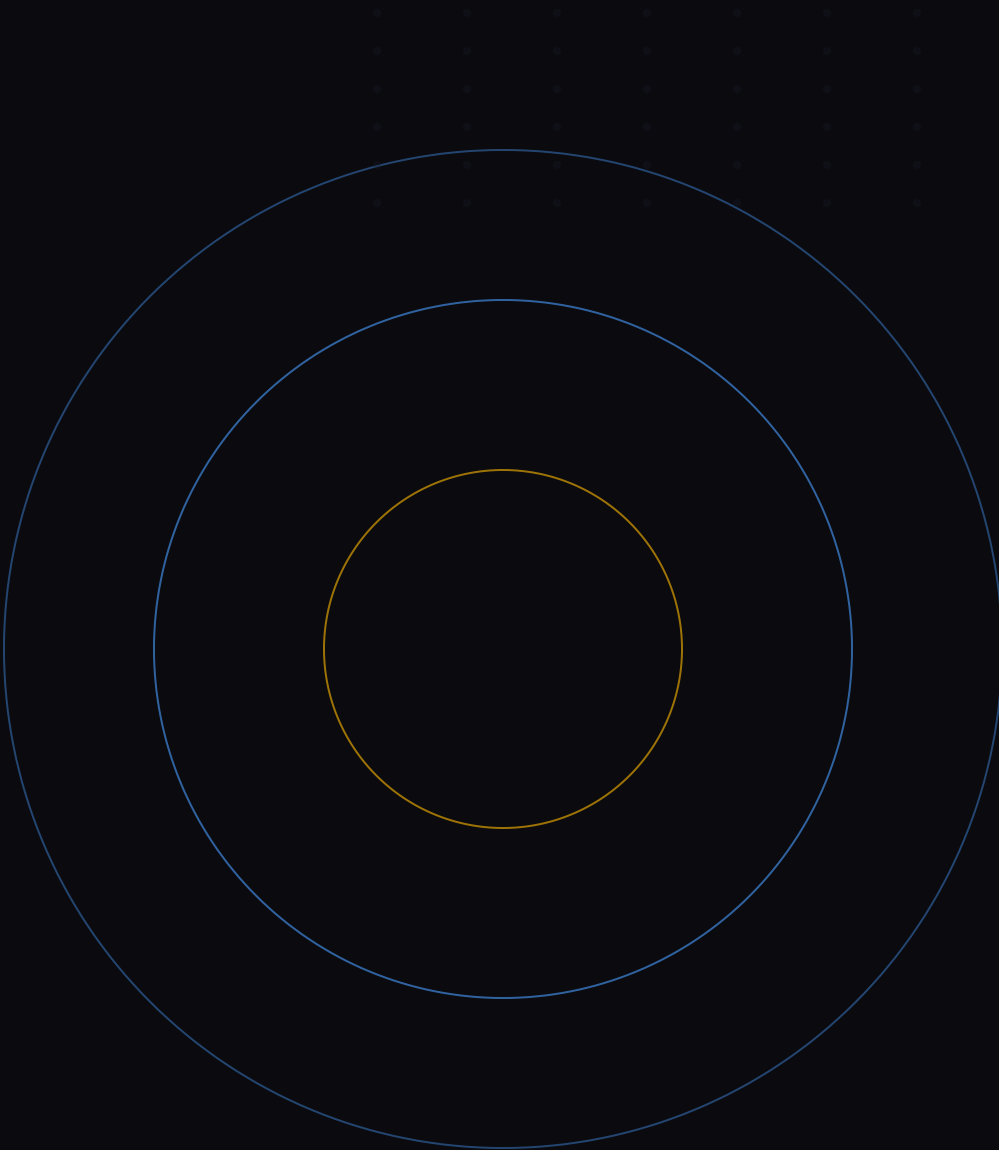


# MASSIF

Predictive Latent Stability Telemetry  
for Autoregressive AI Systems

*Real-Time Latent Stability Monitoring · Dynamical Diagnostics for AI Systems*

DANIEL SOLIS · PRINCIPAL INVESTIGATOR



# Modern frontier models are effectively opaque dynamical systems.

Current AI observability is shallow.

TODAY

Frontier labs ship models with billions of parameters and zero empirical instruments for latent dynamical monitoring.

THE GAP

Interpretability work focuses on factual recall and circuits. The internal dynamical trajectory - where failure actually originates - is unmeasured.

THE RISK

Without measurement of latent dynamics, stability claims are unverifiable. Regulators, insurers, and labs all need a number.



# Recursive stress induces measurable latent stability transitions.

- Persistence  $I_t$  — directional memory of latent velocity
- Tension  $\sigma_t$  — cross-run variance, a temporal sequence
- Volatility Gradient  $\Phi_t$  — stress acceleration,  $d\sigma/dt$
- Predictive Stress Velocity  $\Psi_t$  — smoothed stress momentum
- Curvature  $\kappa_t$  — angular deviation from stable orbit

*"Instability leaves precursor signatures before degeneration occurs."*

Tension rises 2-3 tokens before persistence flips. AUROC > 0.9 across all architectures.

LATENT  
TRAJECTORY · GPT-2  
SMALL  
RECURSIVE  
STRESS  
PROTOCOL ·  
D = 9



# The MASSIF Observability Stack

$$\text{MASSIF} = \kappa \cdot \log_{10}\left( \frac{(\alpha \cdot S)}{(\beta \cdot P)} \times \gamma^D \right)$$

Coefficients:  $\kappa=1.63$ ,  $\alpha=0.97$ ,  $\beta=0.83$ ,  $\gamma=1.17$  (calibrated across 5 architectures)

|    |            |                            |                                                                  |
|----|------------|----------------------------|------------------------------------------------------------------|
| L1 | $I_t$      | Persistence                | Directional stability indicator                                  |
| L2 | $\sigma_t$ | Tension                    | Latent stress - the predictor                                    |
| L3 | $\Phi_t$   | Volatility Gradient        | Stress acceleration                                              |
| L4 | $\Psi_t$   | Predictive Stress Velocity | Smoothed momentum; $\Psi > 0.04$ triggers warning                |
| L5 | $K_t$      | Curvature                  | Trajectory geometry; $\kappa < 0.2$ indicates curvature collapse |
| L6 | $\zeta$    | Recovery Damping           | Post-perturbation return to equilibrium                          |

# MASSIF predicts instability before visible failure.

2-3

TOKENS

>0.9

AUROC ACROSS ARCHITECTURES

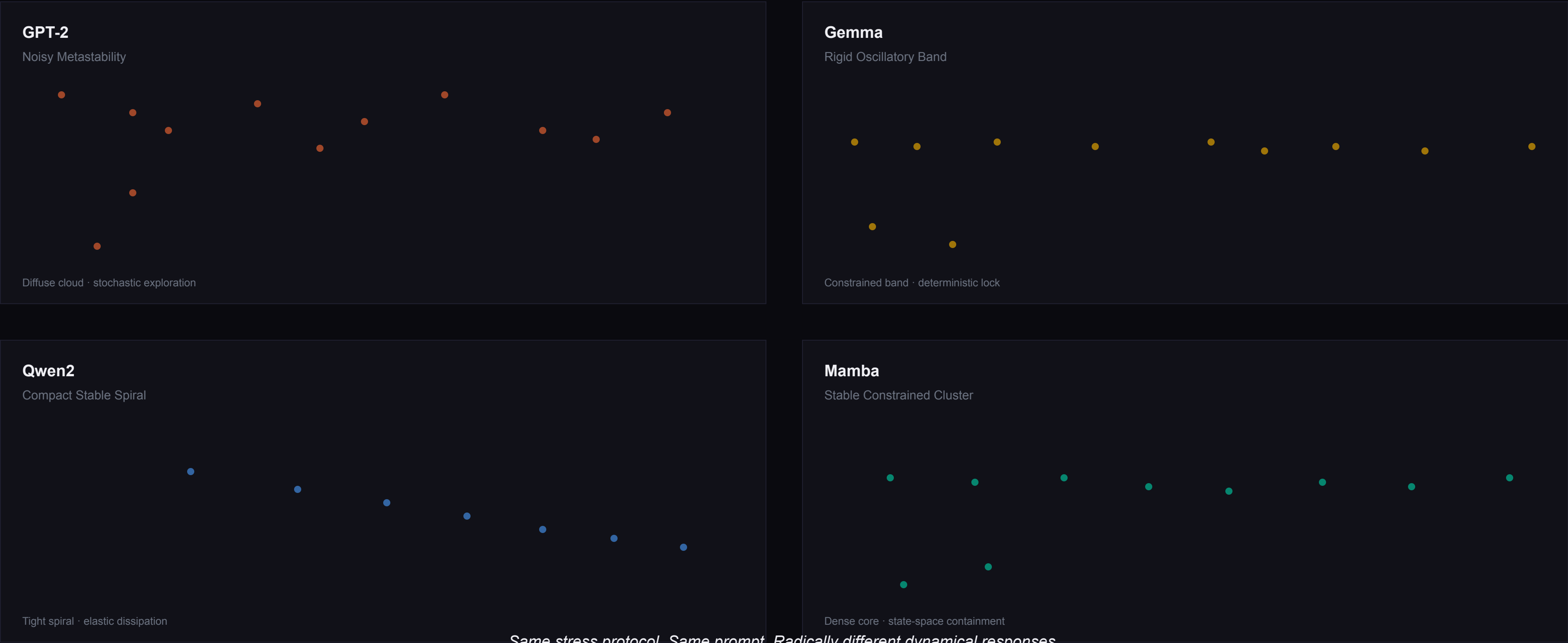
<5%

FPR ON NEUTRAL PROMPTS

| Architecture | Condition  | AUROC | 95% CI         | Lead Time  |
|--------------|------------|-------|----------------|------------|
| GPT-2        | n=4, T=0.5 | 1.000 | [1.000, 1.000] | 2-3 tokens |
| Gemma        | n=4, T=0.5 | 1.000 | [1.000, 1.000] | 2-3 tokens |
| RWKV         | n=2, T=0.5 | 0.994 | [0.977, 1.000] | 2-3 tokens |
| Qwen2        | n=4, T=0.5 | 0.918 | [0.694, 1.000] | 2-3 tokens |

"We do not detect failure. We detect the conditions that produce failure - before they fully form."

# Different architectures occupy distinct latent dynamical regimes.



# Not all architectures respond to stress the same way.

### BRITTLE SYSTEMS

- Accumulate latent stress
- Enter attractor lock-in
- Destabilize under repetition

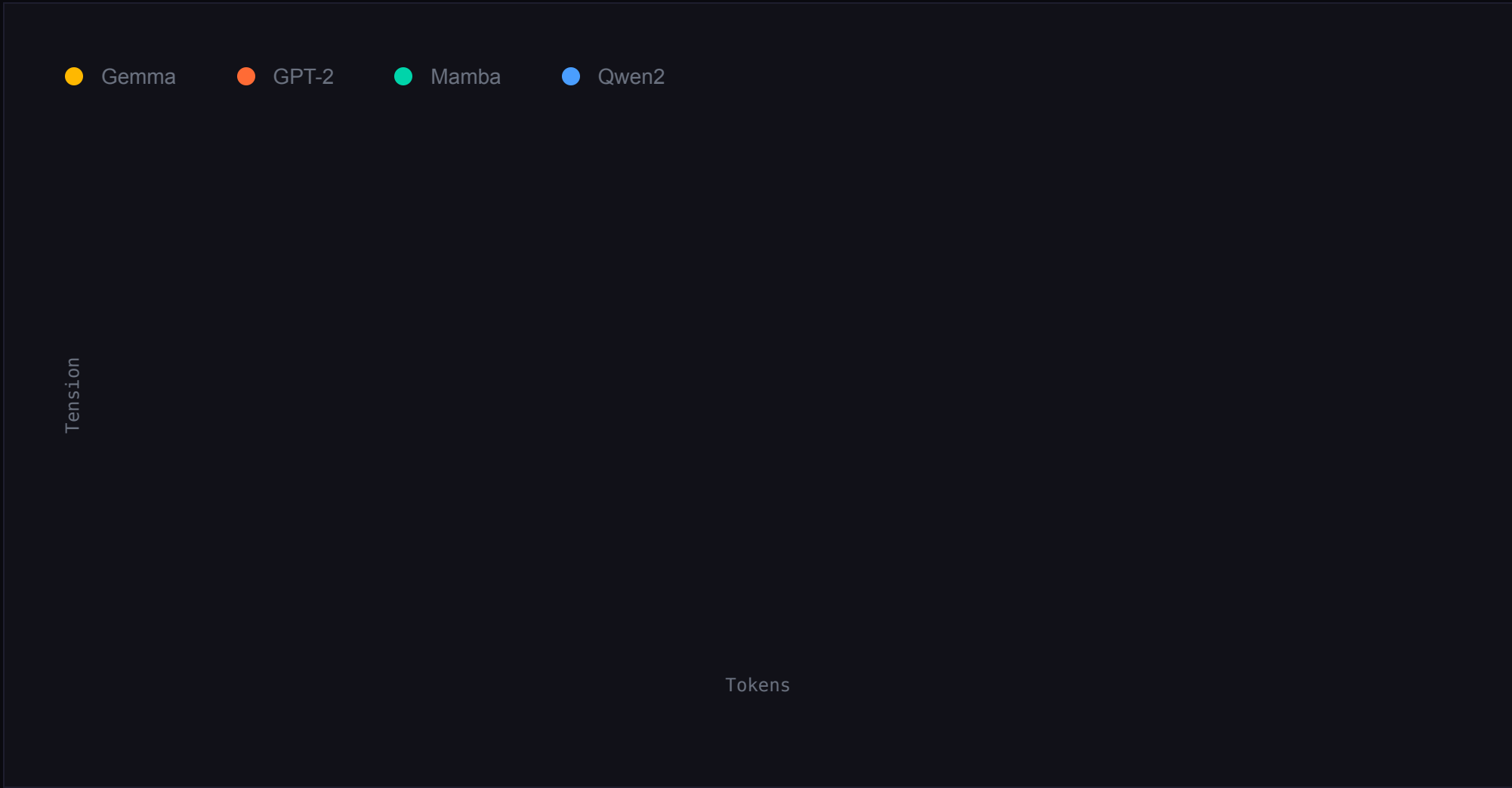
### ELASTIC SYSTEMS

- Dissipate latent stress
- Recover before lock-in occurs
- Maintain dynamical stability

| Class                 | Models       | Mechanism                                     | Flip Rate (n=4, T=0.5) |
|-----------------------|--------------|-----------------------------------------------|------------------------|
| Elastic / Dissipative | Qwen2, Mamba | Stress dissipation                            | 0-10%                  |
| Oscillatory / Brittle | GPT-2, RWKV  | Stress accumulation with oscillatory recovery | 58-70%                 |
| Hysteretic / Rigid    | Gemma        | Rigid lock → sudden destabilization           | 87.5%                  |

Robust architectures do not avoid latent stress - they dissipate it before attractor lock-in occurs.

# Recovery behavior may be more fundamental than flips themselves.



DAMPING ANALYSIS

| Model | $\zeta$ | Regime        |
|-------|---------|---------------|
| Gemma | 0.216   | Underdamped   |
| GPT-2 | 0.579   | Underdamped   |
| Mamba | 0.792   | Overdamped    |
| Qwen2 | 1.187   | Near-critical |

Critical insight: The way a system returns to equilibrium after perturbation reveals its architectural robustness. Recovery dynamics are a fingerprint.

# A candidate observability layer for advanced AI systems.

**Frontier Model Observability**  
Real-time latent monitoring during inference

**Agentic Runtime Monitoring**  
Detect instability precursors in autonomous systems

**Instability Forecasting**  
2-3 token lead time for intervention

**Attractor Detection**  
Identify lock-in before output degeneration

**Online Telemetry**  
Streaming latent instrumentation at scale

**Autonomous Intervention**  
Trigger circuit-breakers on  $\Psi > 0.04$

**Latent Diagnostics**  
Architecture comparison and stress testing

**What MASSIF is:** A predictive latent telemetry framework. **What MASSIF is not:** A full alignment solution or AGI safety guarantee.

We provide measurement. Measurement enables engineering.

# Scaling predictive latent telemetry to frontier systems.

12-MONTH PRIORITIES

- Scale to 7B-70B parameters

and instruction-tuned variants
- Remote telemetry

distributed observability for deployed models
- Streaming latent instrumentation

real-time dashboards for production
- Adaptive intervention

autonomous circuit-breaking based on precursor signals
- Live anomaly detection

unsupervised identification of novel instability modes

ARCHITECTURE COVERAGE

- Transformers
- RWKV
- Mamba
- xLSTM
- Future Architectures

OPEN-SOURCE RELEASE: MASSIF v1

15% of funding allocated to public release

*"The future of AI safety may require  
not only alignment of outputs -  
but observability of latent dynamics."*

# MASSIF

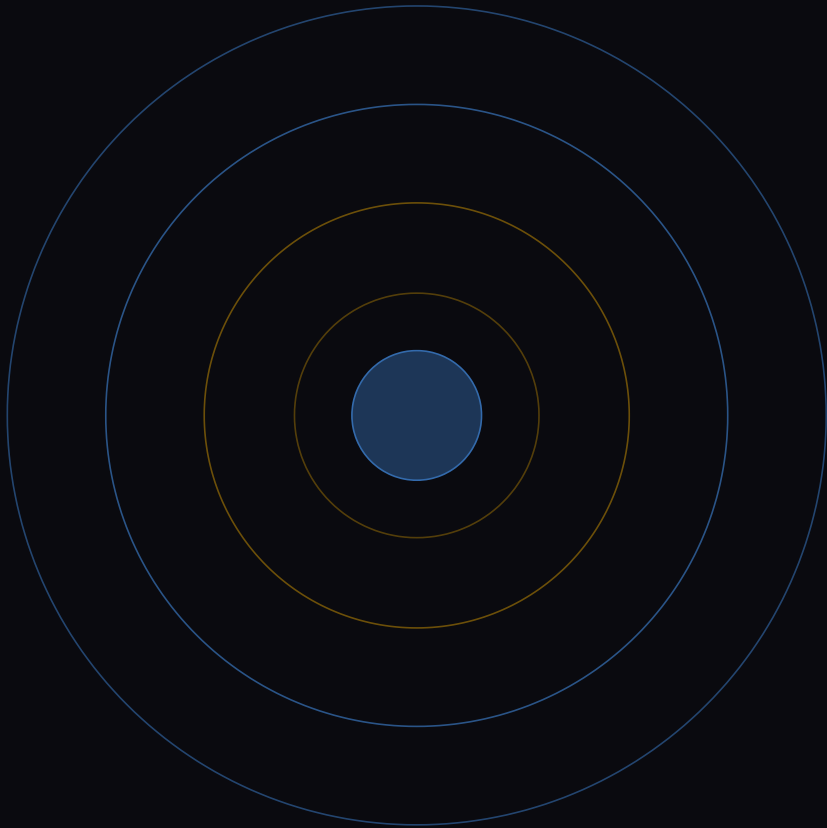
Multiscale Attractor Stability & Stress Inference Framework

Predictive Latent Stability Telemetry for Autoregressive AI Systems

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